

DR. DANIEL RICHARDS

British Antarctic Survey, Cambridge, UK — dancha@bas.ac.uk

PROFESSIONAL APPOINTMENTS

Ice Fracture Modeller (Post-doc) British Antarctic Survey, Cambridge, UK Working with Dr. O. Marsh and Dr. R. Arthern	11/24 - present
Visiting Postdoctoral Research Associate Monash University, School of Earth, Atmosphere and Environment Hosted by Dr. F. McCormack	11/23 - 11/24
Postdoctoral Research Associate University of Tasmania, Institute for Marine & Antarctic Studies Advised by Prof. E. Mantelli	03/23 - 11/24
Postdoctoral Research Associate University of Leeds, School of Mathematics Advised by Dr. S. Pegler & Prof. S. Piazzolo	11/21 - 12/22
Junior Aerodynamicist Force India Formula 1 Team Responsible for part of the design of the front wing, including computational and experimental testing	08/16 - 04/17

EDUCATION

PhD, Fluid Dynamics, University of Leeds Thesis title: <i>Modelling Ice Fabric Evolution and its effect on Viscous Anisotropy</i> Supervised by Dr. S. Pegler, Prof. S. Piazzolo, Dr. O.G. Harlen	2017-21
MSc, Fluid Dynamics (integrated with PhD) University of Leeds (1st class) Dissertation: <i>The Fluid Mechanics of Water Separation from Biodiesel via Coalescence Filtration</i> Supervised by Prof. N. Kapur & Dr. M. Wilson	2017-18
MEng, Aerospace and Aerothermal Engineering, University of Cambridge (1st class) Dissertation: <i>Computing Complex Vortex Flows using Spectral Methods</i> Supervised by Prof. J. Harvey	2012-16

HONORS & AWARDS

UK National Fellowship in Fluid Dynamics (declined) Title: <i>A unified continuum framework for modelling anisotropic polycrystalline flow</i> , a 3 year personal fellowship at the University of Leeds, total amount: £300k	2022
EPSRC mathematical sciences small grant (named researcher) (declined) Title: <i>A general continuum theory of polycrystalline materials</i> , extending the ideas in my PhD from ice to other polycrystalline materials, total amount: £80k	2022
EPSRC Doctoral Prize 1 year fellowship awarded to continue the research from my PhD, University of Leeds	2021
EPSRC Fluid Dynamics CDT full studentship award, £80k	2017-21

HIGHLIGHTED TALKS AND SEMINARS

- Invited talk, AGU23** 2023
In the session *From Grain to Earth: Understanding Plate Tectonics Through Rock Microphysics and Recrystallization to Mantle-Scale Analyses*
- Invited talk, Industrial and Applied Mathematics Seminar, University of Oxford** 2024
- Invited talk, Tübingen Structural Geology-Glaciology seminar** 2022

MENTORING & TEACHING

- Teaching Assistant:** Elementary Differential Calculus MATH0111 2021
School of Mathematics, University of Leeds
- Teaching Assistant:** Modelling with Differential Equations MATH1400 2019
School of Mathematics, University of Leeds
- MSc Student Mentoring:** Maisie Muir, Data Science & Analytics 2021
University of Leeds. I contributed by providing support and guidance to the student in terms of methodology, and contribute to the scientific objectives.

SERVICE TO THE COMMUNITY

- Journal Referee:** Nature Communications, Journal of Geophysical Research, The Cryosphere 2022 - present
- Co-organizer:** 'Maths on Ice' 2023 - 25
An international online seminar series/discussion forum on mathematical glaciology, with Kasia Warburton and Alex Bradley
- Session Convener:** Viscous Flow in Polycrystalline Materials: across disciplines, EGU24, 25, 26 2024-2026

SOFTWARE OUTPUTS

- SpecCAF:** Python implementation of a Spectral Continuum Anisotropic Fabric model, link 2022
- Kraken:** Phase-field poro-visco-elastic model for calving and fracture in ice , link 2026

REFERENCES

- Dr. Robert Arthern** rart@bas.ac.uk
Postdoctoral advisor, British Antarctic Survey
- Prof. Elisa Mantelli** elisa.mantelli@lmu.de
Postdoctoral advisor, University of Tasmania and Ludwig Maximilian University of Munich
- Dr. Sam Pegler** s.pegler@leeds.ac.uk
PhD supervisor, University of Leeds